

Empirical and Behavioral-Simulation Based Evidence for Portfolio-Style Allocation in Student Study Time

Harihar Rengan¹  and Ali-Mansur Valiyev¹ 

Dubai College, Dubai, United Arab Emirates
harihar.rengan@gmail.com, alimansur2009dubai@gmail.com

Abstract. We frame student study time as investment under uncertainty: a limited time budget is divided across subjects whose grades rise through capped, diminishing-returns curves. We compare three allocation policies—uniform, greedy, and portfolio—across three experiments. On EdNet KT3 ($\sim 297,000$ real learners), portfolio allocators best reproduce observed behavior, and the gap to greedy widens for stronger students. In a simulation of 8,000 students carrying 11 behavioral-economics traits, each grounded in published psychology and decision-theory estimates and coupled through a Gaussian copula, a portfolio $>$ uniform $>$ greedy GPA ordering holds across every skill quintile under fixed effort, while allowing effort to vary hands dominance to skill and gross effort; student self-correction is the mechanism that naturally removes greedy allocation from cohorts. In a reinforcement-learning experiment, a policy-discovery agent that observes only noisy weekly performance and never the ground-truth learning rate independently converges on diversified allocation, indicating the result does not depend on knowing true ability. Throughout, the decisive contrast is between diversified and concentrated allocation: uniform is the equal-weight $1/N$ portfolio and captures $\sim 86\%$ of the GPA gain over greedy, with optimization contributing the remainder. We propose portfolio allocation as an effective policy for allocating students’ time, and note two practical consequences: allocation coaching is most valuable for learners who cannot easily self-correct, and institutional scheduling that affords frequent self-evaluation moves cohorts toward diversified allocation without individual instruction.

Keywords: study-time allocation · behavioral simulation · portfolio choice · self-regulated learning · Gaussian copula · knowledge tracing · reinforcement learning

1 Introduction

Students and schools increasingly emphasize the transferable skill of time allocation, yet students struggle with it amid distractions, fluctuating motivation, and

Both authors contributed equally to this work.

indecisiveness. A student has a limited time budget that must be divided among subjects, akin to dividing investment between assets with uncertain returns.

The problem itself has been studied from various angles, and some existing answers raise concerns. Literature in psychology studies this directly as study-time allocation and finds that when time is limited, such as just prior to a test, learners spend it on the items they judge easy rather than hard [1]; the region-of-proximal-learning model refines this to a learner who abandons what they believe they know and dwells on middling difficulty [2]. Both describe someone concentrating on the strength of a belief about their own ability—that is greedy allocation, which we have represented and tested algorithmically. Self-regulated-learning research treats such planning as a teachable skill rather than a fixed disposition [3], making study-time allocation one of the determinants of achievement a student can actually be taught to improve. Yet neither the scheduling literature, which shows that distributing study over time changes what is retained [4], nor knowledge tracing, which estimates what a learner knows and how fast they learn [5,6], addresses the allocation policy that maximizes results. Consequently, we bridge this gap through computational analysis and simulation.

We supply that rule by taking the investment analogy seriously, applying the Markowitz mean-variance optimizer [8] and Black-Litterman [9] and comparing them to naive uniform allocation and greedy, strength-concentrated allocation. Note that uniform allocation is the $1/N$ form of portfolio allocation. Portfolio theory already anticipates that $1/N$ is hard to beat when the inputs must be estimated: optimizing an allocation requires an estimate of each subject’s pay-off, and those estimates are noisy, so the error they introduce tends to cancel the advantage that optimization would hold under perfect information. This is not merely theoretical—when fourteen optimization methods were each tested against the simple equal-split rule on seven real datasets, none consistently beat it out of sample [10]. The student-time allocation problem is situated in that regime, since students can never observe their true per-subject learning rates. We therefore read our policies as two diversified allocators ($1/N$ and optimized) against one concentrated allocator (greedy), and report the $1/N$ comparison as a first-class result. Our contributions are: (i) an empirical test on EdNet of how closely real behavior resembles each policy; (ii) three literature-grounded trait-sampling regimes and a counterfactual simulation isolating allocation from effort and skill; and (iii) a reinforcement-learning account of the traits an outcome-maximizer actually rewards.

2 Method

2.1 Empirical Evaluation on EdNet KT3

EdNet records timestamped interactions across seven TOEIC knowledge parts, serving as per-skill ability signals and objective time records independent of self-reporting [11]. We use an 80–20 train-test split at the student level and draw four stratified out-of-sample cohorts by final-week performance and week-over-week

trajectory: high performers (top of the final-score distribution), low performers (bottom), improvers (grade slope > 0.05 over the term), and the general held-out set (43/243/36/286 eligible learners).

The model has two layers. A **belief layer** takes prior study time, its allocation, and grades from week 1 to $T-1$ and predicts week- T per-subject grades. We use a Bayesian Hierarchical Model (BHM) emitting per-skill learning rates $\hat{\theta}_j$ from which grades follow via the mastery curve

$$\hat{g}_j = 0.25 + 0.75(1 - e^{-\hat{\theta}_j H_j}), \quad (1)$$

where H_j is cumulative hours on skill j and 0.25 is the guessing floor; we also try a Deep Knowledge Tracing LSTM [6] and a Multi-Layer Perceptron (MLP). An **allocator layer** receives the raw weekly hours and the per-subject grades to match, and reproduces them via its optimizer; returned hours are compared to observed hours by RMSE. Here the optimizer is asked to match a cohort’s grades rather than optimize them, so this measures how closely each group resembles each policy—resemblance, not benefit. Absolute hours are essential: renormalizing to study probabilities discards effort and inverts the ranking, so KL divergence crowns greedy where hours-RMSE ranks portfolio first; we therefore report hours-RMSE throughout.

2.2 Trait-Grounded Simulation

Each agent carries eleven behavioral-economics traits drawn from published marginals (Table 1); as no source supplies a full distribution, each is cited for its construct or central tendency and every dispersion parameter is a calibration choice. Each subject also receives a decoupled innate learning rate $\theta_j \sim \text{LogNormal}$, grounded in per-skill heterogeneity from knowledge tracing [5,7]. Six cited inter-trait correlations (Table 2), split into a consensus meta-analytic pair (conscientiousness–self-efficacy, $\rho = 0.36$ [12]) and five contested pairs, are imposed through a Gaussian copula: we project the target matrix to the nearest valid correlation matrix (Higham), retain eigen-components with eigenvalue > 1 , and map each latent through the normal CDF and the inverse marginal (Sklar’s construction), hitting every marginal exactly while a Cholesky cross-check recovers each ρ to within 0.01. We run three regimes—independent, consensus-only, and all-cited—and, since ability is sampled outside the copula, allocation style and ability stay separable.

Each agent faces $J = 6$ skills over 70 study-days; hours convert to mastery through Eq. (1), and a 20-item-per-skill exam yields per-subject scores whose mean is GPA. Under *fixed* effort each student receives 5 h/day; under *variable* effort, sleep capacity and a trait-driven tendency to waste time let total effort float. Agents never observe true θ : each holds a belief $\hat{\theta}_j$, tests itself daily on noisy quizzes filtered by immutable traits (overconfidence inflates, conscientiousness sharpens), and updates that belief (EWMA) before re-allocating. We sweep the number of self-evaluations from 70 down to one. The grade-optimal (portfolio) allocation is found by bisection on the marginal-return threshold; greedy is $h \propto \theta$;

Table 1: The eleven trait marginals and the per-skill ability rate, with the source grounding each. **G**: central value literature-grounded; **C**: construct cited, distribution a calibration choice.

trait	marginal	grade	source
β present-bias	$\text{Beta}(8, 1.2) \rightarrow [0.4, 1]$	G	Augenblick et al. 2015 [13]
δ discount	$\text{Beta}(20, 5)^{1/52}$	C	Andersen et al. 2008 [14]
λ loss-aversion	$\text{LogN}(\ln 1.95, 0.35)$	G	Brown et al. 2024 [15]
γ CRRA curvature	$\text{TruncN}(0.72, 0.4^2)$	C/G	Holt & Laury 2002 [16]
η planning-fallacy	$\text{LogN}(\ln 0.62, 0.25)$	G	Buehler et al. 1994 [17]
α overconfidence	$\text{N}(0.15, 0.2^2)$	C	Moore & Healy 2008 [18]
c conscientiousness	$\text{N}(0, 1) \rightarrow [0, 1]$	C	Soto 2016 [19]
s self-efficacy	$\text{N}(0, 1) \rightarrow [0, 1]$	C	Bandura 1997 [20]
φ peer-weight	$\text{Beta}(2, 8) \cdot 0.4$	G	Sacerdote 2001 [21]
ψ sleep capacity	$\text{N}(1, 0.15^2)$	G	Carrell et al. 2011 [22]
k mental-accounting	$\text{Uniform}\{1..4\}$	C	Thaler 1985 [23]
θ_j ability (per skill)	$\text{LogNormal, decoupled}$	C	Corbett & Anderson 1995 [5]

Table 2: The six cited inter-trait correlations (symbols as in Table 1). Category A (wide consensus) is one meta-analytic pair; Category B (present but contested) supplies the other five, each swept to zero for robustness.

pair	ρ	category	basis
$c-s$	+0.36	A	meta-analytic [12]
$\alpha-\eta$	+0.40	B	overconfidence / planning [17,18]
$\alpha-s$	+0.30	B	shared positive self-belief
$\beta-c$	+0.25	B	self-control
$\beta-\eta$	-0.20	B	patience aids time estimates
$\lambda-\gamma$	+0.30	B	mixed risk elicitations

uniform splits evenly. To rank trait importance we train an MLP (two hidden layers 32,16, early stopping, $N = 8000$) to predict GPA from traits, scored by R^2 -drop permutation importance.

2.3 Reinforcement Learning

In the same environment, a policy-discovery agent (REINFORCE [24]) sees only noisy weekly quiz scores and cumulative hours—never θ —emits a weekly 6-simplex allocation via a Dirichlet policy, and is rewarded only by terminal GPA, testing whether a reward-maximizer discovers portfolio allocation with no reference supervision. A separate trait-discovery search (differential evolution and a natural-evolution-strategy gradient) treats the 11 traits as mutable and, with a gradient-boosting surrogate, reports which traits the reward actually rewards.

Table 3: Experiment 1. Hours-RMSE (lower = closer to real behavior) by belief×allocator on the four EdNet cohorts. Bold marks the best per cohort; greedy is consistently among the worst. Degenerate rows omitted.

combo	low	general	improver	high
BHM+Markowitz	0.175	0.186	0.198	0.244
BHM+Black–Litterman	0.176	0.186	0.198	0.244
BHM+Uniform	0.177	0.187	0.198	0.246
BHM+Greedy	0.191	0.203	0.247	0.273
MLP+Markowitz	0.197	0.203	0.217	0.241

3 Results

3.1 Experiment 1 — EdNet KT3

Portfolio allocators consistently beat greedy and either beat or tie uniform, by a smaller margin (Table 3). Holding the belief layer fixed, uniform—the $1/N$ portfolio—sits within 0.002 hours-RMSE of the optimized allocator in every cohort, while greedy trails by roughly an order of magnitude more; the separation is between diversified and concentrated allocation, not among the diversified policies. BHMs are the best belief layer and Markowitz the best allocator. The greedy–portfolio gap is larger for higher-performing and improving students (0.033 and 0.049 hours-RMSE versus 0.015 and 0.018 for low and general cohorts), reflecting that stronger students already resemble the portfolio policy. This is the empirical anchor on real data.

3.2 Experiment 2 — Trait-Grounded Simulation

The correlation regime was nearly irrelevant to the outcome. Both under fixed effort and variable effort, GPA rises monotonically greedy $\rightarrow$ uniform $\rightarrow$ portfolio across all five skill quintiles, and the benefit grows with skill (Table 4). Because every agent is evaluated under all three policies, the contrasts are paired; Table 5 reports them with bootstrap confidence intervals and effect sizes. Under fixed effort, uniform recovers 5.89 of the 6.81 GPA points separating greedy from the optimized portfolio (86.4%), and against greedy both diversified policies land a large effect (Cohen’s $d_z = 1.19$ and 1.09) where against each other they do not ($d_z = 0.21$). We do not lean on p -values, since $N = 8000$ is a chosen sample size and any true difference can be driven to arbitrary significance by running more agents; the intervals and effect sizes are the honest quantities. Avoiding concentration is the first-order decision, optimizing weights the second-order one, and the $1/N$ share holds within three points across every quintile.

Agents nonetheless approach the portfolio optimum through daily self-correction, by approaching that very allocation style. Grade loss from misallocation falls from ~ 3.1 GPA points under a single evaluation to ~ 0.45 under daily evaluation, while allocation RMSE to the portfolio optimum falls $0.064 \rightarrow 0.027$ and to uniform $0.053 \rightarrow 0.035$ —closer to both, but markedly more to portfolio (Fig. 1)—while distance to greedy stays pinned at ~ 0.11 . Consequently, at

Table 4: Experiment 2. Counterfactual GPA by innate-skill quintile (Q1 weakest $\rightarrow$ Q5 strongest) under both effort regimes; Δ = portfolio – greedy. The portfolio $>$ uniform $>$ greedy ordering holds in every quintile, and the advantage of diversified allocation grows with skill.

θ quintile	fixed effort				variable effort			
	greedy	uniform	portf.	Δ	greedy	uniform	portf.	Δ
Q1 (weakest)	72.9	77.2	77.9	+5.0	68.3	71.6	72.1	+3.7
Q2	76.5	82.1	83.0	+6.5	72.2	76.6	77.1	+4.9
Q3	78.9	84.8	85.7	+6.8	74.3	79.2	80.0	+5.7
Q4	80.9	87.4	88.4	+7.5	76.8	82.4	83.2	+6.4
Q5 (strongest)	83.9	90.9	92.1	+8.2	79.8	86.3	87.3	+7.5
all	78.6	84.5	85.4	+6.8	74.3	79.2	79.9	+5.6

Table 5: Experiment 2. Paired contrasts on per-student counterfactual GPA ($N=8000$ per regime). Mean difference [95% bootstrap CI, 10,000 resamples] with paired Cohen’s d_z . Final row: the share of the greedy $\rightarrow$ portfolio gap captured by uniform, the $1/N$ portfolio.

contrast	fixed effort		variable effort	
portfolio – greedy	+6.81	[6.69, 6.94], $d_z=1.19$	+5.64	[5.51, 5.77], $d_z=0.94$
uniform – greedy	+5.89	[5.76, 6.00], $d_z=1.09$	+4.92	[4.79, 5.05], $d_z=0.85$
portfolio – uniform	+0.93	[0.83, 1.03], $d_z=0.21$	+0.72	[0.61, 0.83], $d_z=0.15$
$1/N$ share of gap	86.4%	[85.0, 87.7]	87.2%	[85.4, 89.0]

higher self-correction counts effort and skill correlate with GPA far more than allocation style. Our MLP confirms this under R^2 -drop permutation importance: with skill and variable effort available, skill and gross-effort traits dwarf the allocation traits, and a standardized decomposition agrees that effort ($\beta=+4.18$) and skill ($+5.36$) dominate while allocation style is flat ($\beta=-0.06$), the raw $+0.34$ style–GPA correlation being an ability confound (Fig. 2). The allocation traits, most malleable from the start, self-correct away.

3.3 Experiment 3 — Reinforcement Learning

Rewarded only by exam GPA and never shown θ , the REINFORCE policy reaches 87.1 GPA—against an optimal upper bound 89.0, uniform 87.9, greedy 82.5, random 76.8—closing 70% of the greedy $\rightarrow$ optimal gap and beating greedy by $+4.5$ points, with realized allocations scoring 0.94 in closeness to portfolio style. A pure reward-maximizer thus discovers diversified allocation rather than greedy, with no access to true skill. Consistent with Experiments 1 and 2 it does not beat the $1/N$ portfolio at 87.9: what it reliably discovers is diversification, not the optimized weights; trained in the same environment as Experiment 2, this is emergence within our model rather than independent confirmation. The trait-discovery search tells the complementary story: differential evolution

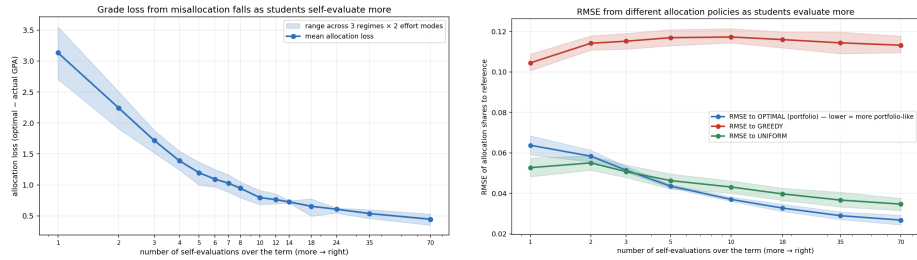


Fig. 1: Self-correction over the term (1 to 70 self-evaluations; band = range across the three correlation regimes $\times$ two effort modes). **Left:** grade loss from misallocation (optimal – actual GPA) falls as students self-evaluate more. **Right:** RMSE of the agent’s allocation shares to the optimal (portfolio), uniform, and greedy references; self-evaluation lowers the distance to both the optimal and uniform references, but markedly more to the portfolio optimum ($0.064 \rightarrow 0.027$) than to uniform ($0.053 \rightarrow 0.035$), while distance to greedy stays high (~ 0.11).

and the NES gradient both reach ~ 92 GPA (population mean 77) on the same profile—high sleep capacity, high conscientiousness, low present-bias—with every allocation-relevant trait near zero (sleep 0.96, conscientiousness 0.35 in R^2 -drop units). The reward rewards traits that buy effort and accurate self-belief, not a fixed allocation style, because the style is something the agent learns to correct on its own.

4 Discussion and Limitations

Framing student time as investment under uncertainty and using portfolio allocators produces the best GPA outcomes we measure, on real data and in a simulation of diverse learning agents. The decisive step, however, is diversification itself rather than its optimization: the $1/N$ portfolio captures most of the gain over greedy, exactly as portfolio theory predicts when returns must be estimated [10]. This matters because greedy is not merely hypothetical—it is what the study-time-allocation literature reports learners reaching for when time is short [1,2]. The policy we find worst is the one students default to, and the remedy fits in a sentence: give every subject an equal share. Because allocation traits are the most changeable, their initial state gives little signal toward final grades, whereas effort and innate skill correlate strongly; yet allocation is precisely the lever that is teachable and transferable, unlike innate skill. Fixing effort through school policy and holding regular condensed testing (exams of all subjects in a short window) affords ample self-evaluation and focuses achievement on the allocation policy.

Several limitations contextualize these claims. The grade function (Eq. (1)) is concave and capped, so diminishing returns are assumed rather than measured; that assumption favors diversification over concentration by construction, and the greedy gaps of Experiment 2 are conditional on it. Experiments 2 and 3 share one environment, so their agreement tests internal consistency, not external va-

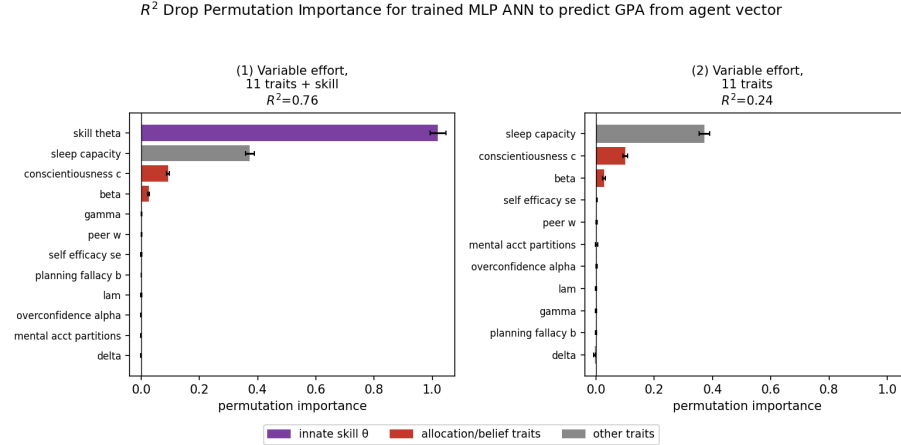


Fig. 2: ANN trait importance for GPA (R^2 -drop permutation importance). Innate skill θ dominates; the only disposition that registers is sleep capacity, an effort driver. Allocation/belief traits are negligible. A third condition (fixed effort, skill withheld) is omitted: there the network has no signal ($R^2 \approx 0$) and every importance is ≈ 0 .

lidity; only Experiment 1 touches real learners, and it measures resemblance rather than outcomes, so we make no causal claim that reallocating a real student’s hours would raise their grades.

5 Conclusion

Across $\sim 297,000$ real EdNet learners, a trait-grounded simulation, and a reinforcement learner blind to true skill, portfolio-style allocation consistently beat greedy and tied or beat uniform, and the portfolio $>$ uniform $>$ greedy ordering held across every skill quintile under fixed effort. The robust finding is that concentration costs students grades: the equal-weight $1/N$ portfolio captures most of what diversification offers, and optimized weights add a real but second-order increment. Allowing effort to vary reduced allocation’s measured importance—not because allocation is cheap, but because self-correction naturally removes greedy allocation from cohorts. The practical implication is that allocation coaching should target learners who cannot easily self-correct and be paired with effort support.

Acknowledgments. We thank EdNet, whose dataset formed our empirical anchor, and the studies cited below from which we drew our trait distributions and correlations. We also sincerely thank Professor David Laibson for his time and key initial advice on our methods. An AI assistant was used solely for copy-editing and formatting of the manuscript; all research design, implementation, analysis, results, and conclusions are the authors’ own.

References

1. Son, L.K., Metcalfe, J.: Metacognitive and control strategies in study-time allocation. *J. Exp. Psychol. Learn. Mem. Cogn.* **26**(1), 204–221 (2000)
2. Metcalfe, J., Kornell, N.: A region of proximal learning model of study time allocation. *J. Mem. Lang.* **52**(4), 463–477 (2005)
3. Zimmerman, B.J.: Becoming a self-regulated learner: an overview. *Theory Pract.* **41**(2), 64–70 (2002)
4. Cepeda, N.J., Pashler, H., Vul, E., Wixted, J.T., Rohrer, D.: Distributed practice in verbal recall tasks: a review and quantitative synthesis. *Psychol. Bull.* **132**(3), 354–380 (2006)
5. Corbett, A.T., Anderson, J.R.: Knowledge tracing: modeling the acquisition of procedural knowledge. *User Model. User-Adapt. Interact.* **4**(4), 253–278 (1995)
6. Piech, C., et al.: Deep knowledge tracing. In: *Adv. Neural Inf. Process. Syst.* (NeurIPS) 28, pp. 505–513 (2015)
7. Koedinger, K.R., Carvalho, P.F., Liu, R., McLaughlin, E.A.: An astonishing regularity in student learning rate. *PNAS* **120**(13), e2221311120 (2023)
8. Markowitz, H.: Portfolio selection. *J. Finance* **7**(1), 77–91 (1952)
9. Black, F., Litterman, R.: Global portfolio optimization. *Financ. Anal. J.* **48**(5), 28–43 (1992)
10. DeMiguel, V., Garlappi, L., Uppal, R.: Optimal versus naive diversification: how inefficient is the $1/N$ portfolio strategy? *Rev. Financ. Stud.* **22**(5), 1915–1953 (2009)
11. Choi, Y., et al.: EdNet: a large-scale hierarchical dataset in education. In: *Proc. AIED 2020*, pp. 69–73 (2020)
12. Barańczuk, U.: The five-factor model of personality and generalized self-efficacy: a meta-analysis. *J. Individ. Differ.* **42**(4), 183–193 (2021)
13. Augenblick, N., Niederle, M., Sprenger, C.: Working over time: dynamic inconsistency in real effort tasks. *Q. J. Econ.* **130**(3), 1067–1115 (2015)
14. Andersen, S., Harrison, G.W., Lau, M.I., Rutström, E.E.: Eliciting risk and time preferences. *Econometrica* **76**(3), 583–618 (2008)
15. Brown, A.L., Imai, T., Vieider, F.M., Camerer, C.F.: Meta-analysis of empirical estimates of loss aversion. *J. Econ. Lit.* **62**(2), 485–516 (2024)
16. Holt, C.A., Laury, S.K.: Risk aversion and incentive effects. *Am. Econ. Rev.* **92**(5), 1644–1655 (2002)
17. Buehler, R., Griffin, D., Ross, M.: Exploring the “planning fallacy”. *J. Pers. Soc. Psychol.* **67**(3), 366–381 (1994)
18. Moore, D.A., Healy, P.J.: The trouble with overconfidence. *Psychol. Rev.* **115**(2), 502–517 (2008)
19. Soto, C.J.: The little six personality dimensions from early childhood to early adulthood. *J. Personality* **84**(4), 409–422 (2016)
20. Bandura, A.: *Self-Efficacy: The Exercise of Control*. W.H. Freeman, New York (1997)
21. Sacerdote, B.: Peer effects with random assignment: results for Dartmouth roommates. *Q. J. Econ.* **116**(2), 681–704 (2001)
22. Carrell, S.E., Maghakian, T., West, J.E.: A’s from Zzzz’s? The causal effect of school start time on the academic achievement of adolescents. *Am. Econ. J. Econ. Policy* **3**(3), 62–81 (2011)
23. Thaler, R.: Mental accounting and consumer choice. *Mark. Sci.* **4**(3), 199–214 (1985)
24. Williams, R.J.: Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Mach. Learn.* **8**(3), 229–256 (1992)